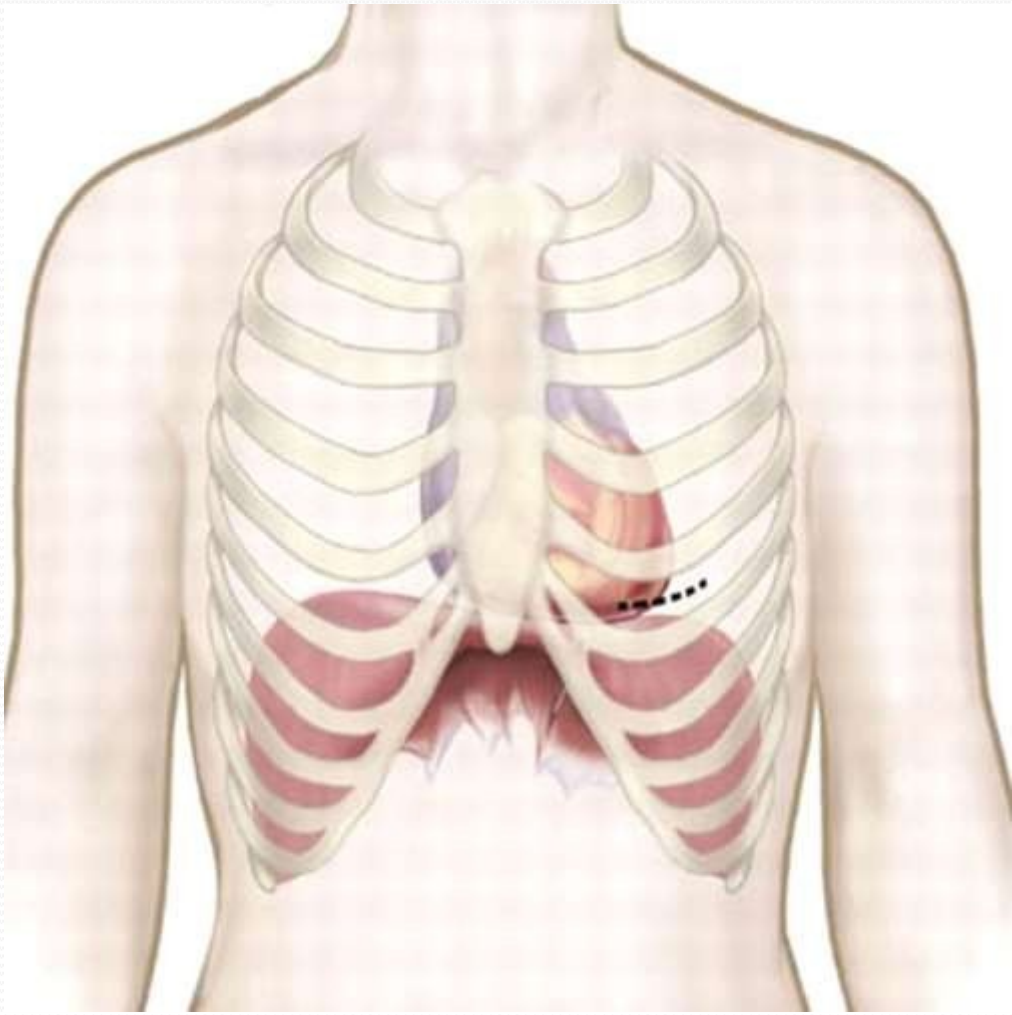

بسم الله الرحمن الرحيم

Intercostal space

Prepared By: Babar Ali



Presentation outlines

- Intercostal space .
- Intercostal muscles & their action.
- Neurovascular supply.
- Importance of intercostals .

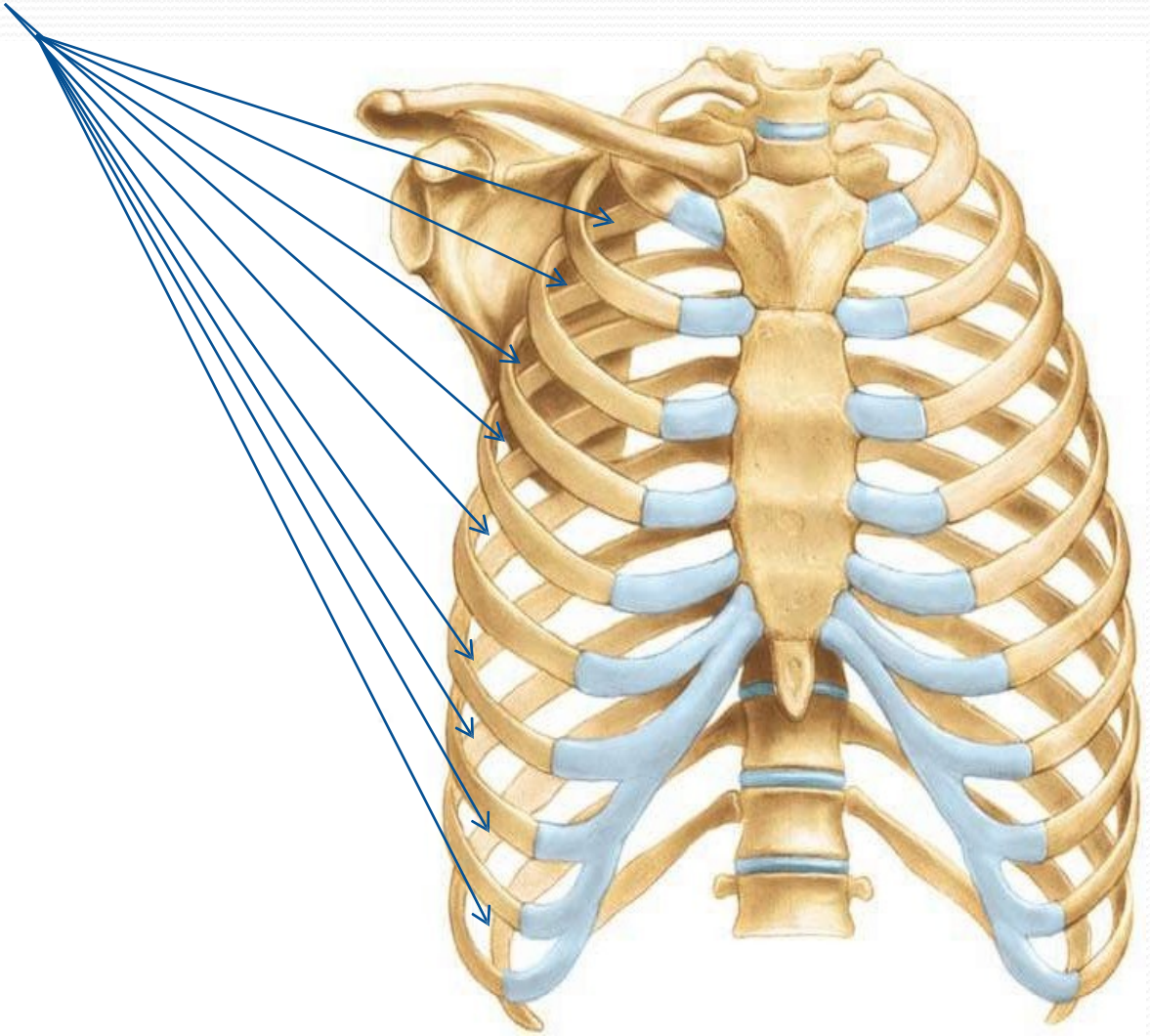
Intercostal space

- The space between two adjacent ribs(costal) is known as intercostal space.
- There are 12th pairs of ribs and 11th pairs of inter costal space at thoracic wall.
- Interiorly inter costal space are lined by different structure at thoracic wall in sequence; Skin-Superficial fascia-Serratus anterior muscle and in between by intercostal muscles.
- Posterioerly by endothoracic facsia-Perital pleura-vicsral pleura
and lungs.

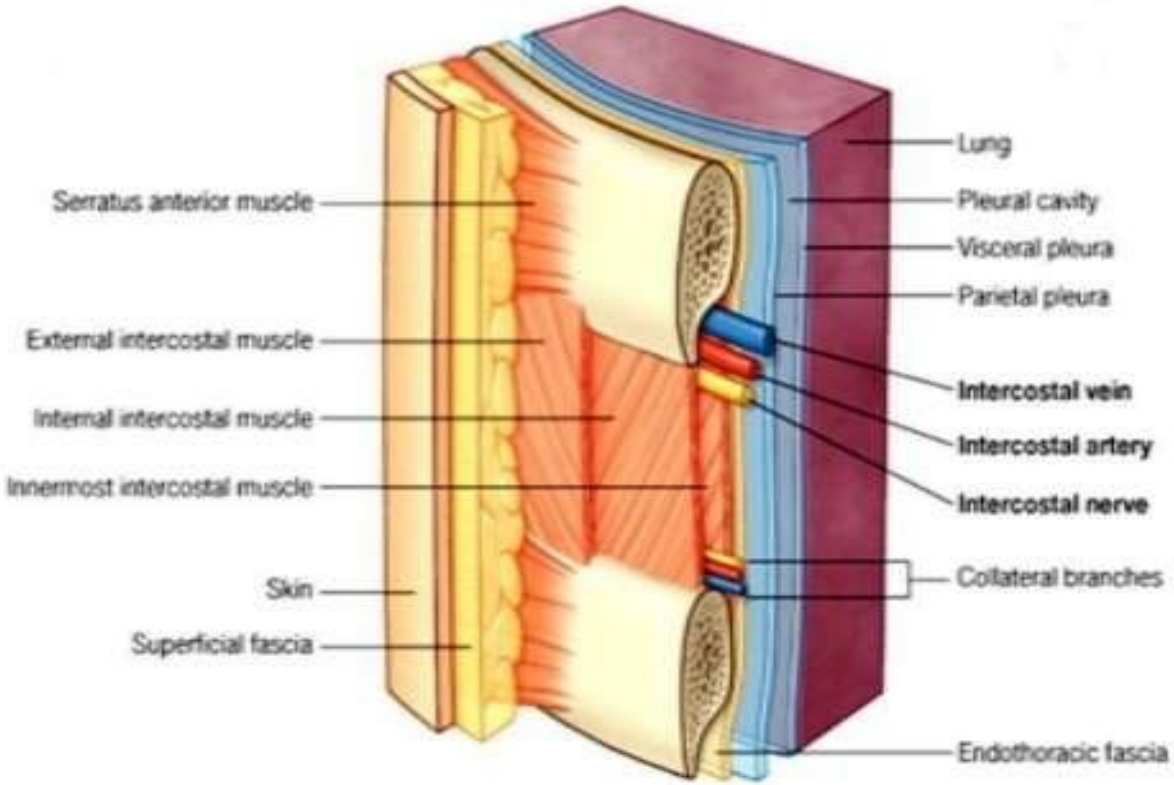
Note: Costal spaces are counted by palpating starnal angle.

Intercostal

s



Contents of Intercostal space



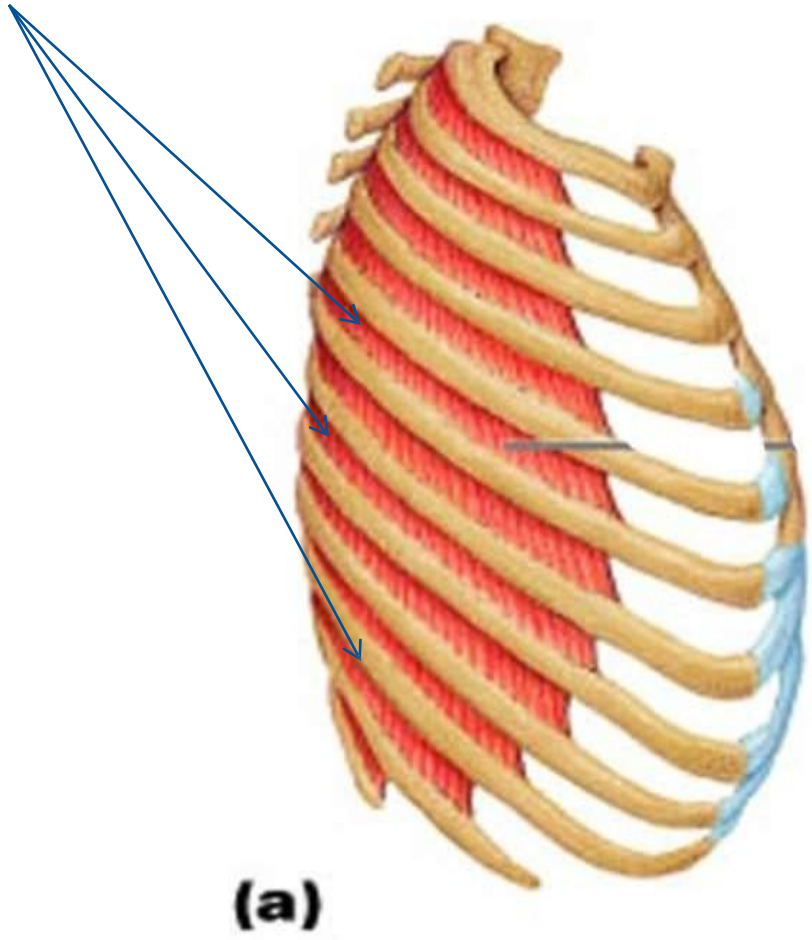
Intercostal muscle

- The muscles which cover the spaces in between adjacent ribs(intercostal space) known as inter costal muscles.
- There are three types of intrecostal muscles namely;
  - External intercostal muscle.
  - Internal inter costal muscle.
  - Inermost intercostal muscle.

External intercostal muscle

- External intercostals (11 pairs) these outermost muscles pass from the lower border of the upper rib to the upper border of the lower rib downword and forward. They extend anteriorly towards the costochondral junction becoming gradually more fibrous, and merge into the external(anterior) intercostal membrane.

External intercostal muscl



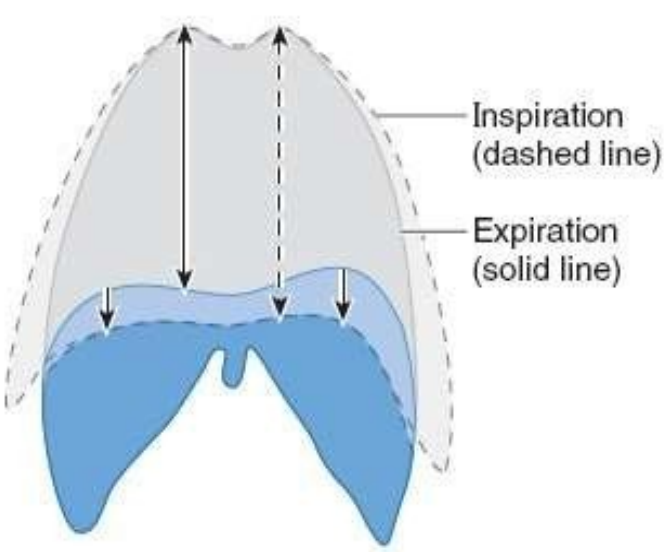
Actio

- During inspiration these muscle contract and pull up the ribs to increase surface area for lungs to inspire.

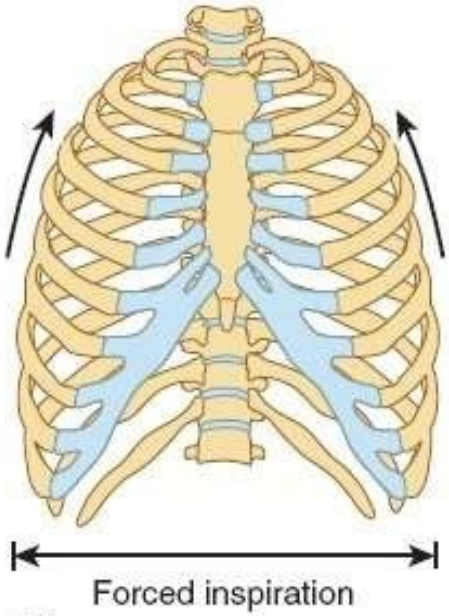


Anteroposterior Expansion

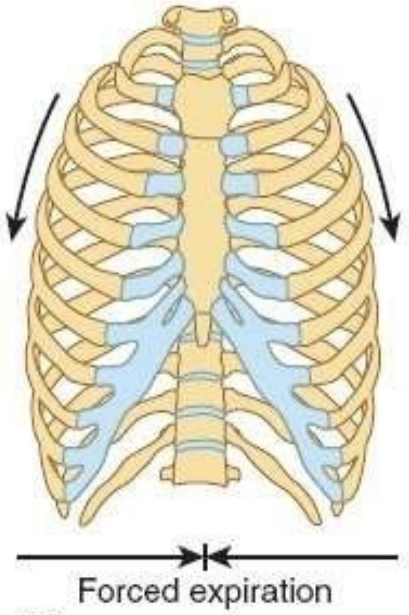
Schematic view



(A) Combination of rib movements



(B)



(C)

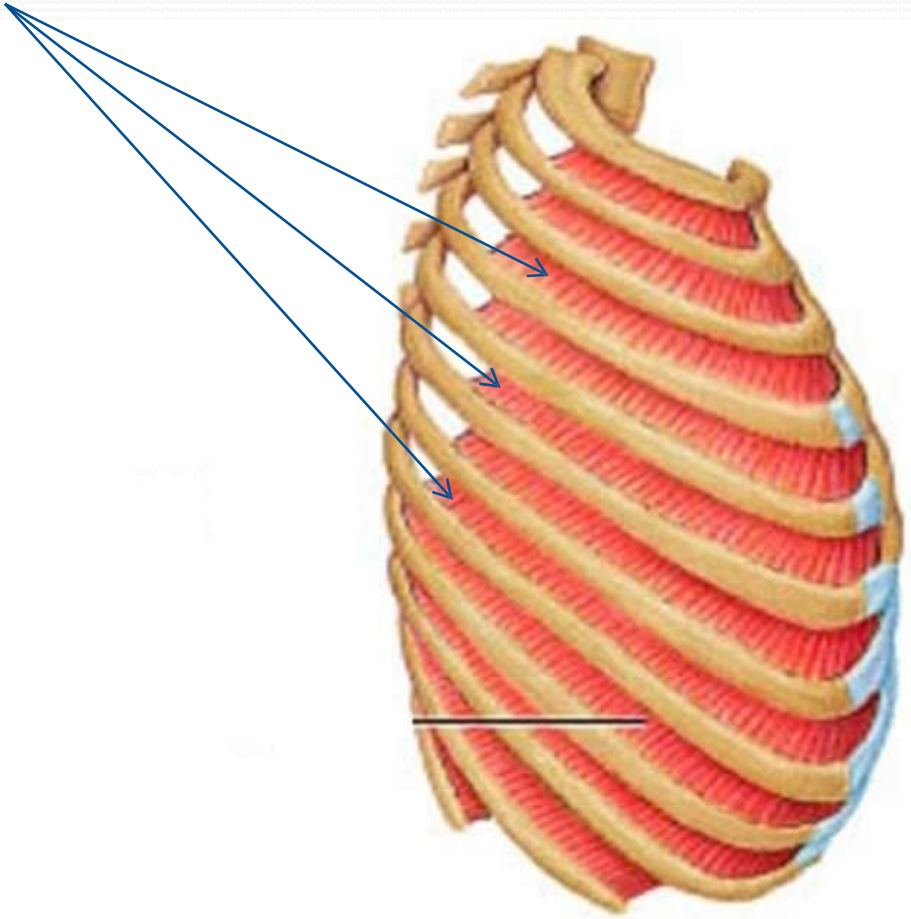
Anterior views

By Babar ali

Internal intercostal muscle

- Internal intercostals (11 pairs) run intermediately , and with the fibres directed downward and backward running at right angles, to the external intercostals. They extend from the sternum, laterally around to the angle of the rib, where they become fibrous and merge to form the internal (posterior) intercostal membrane.

Internal intercostal musc



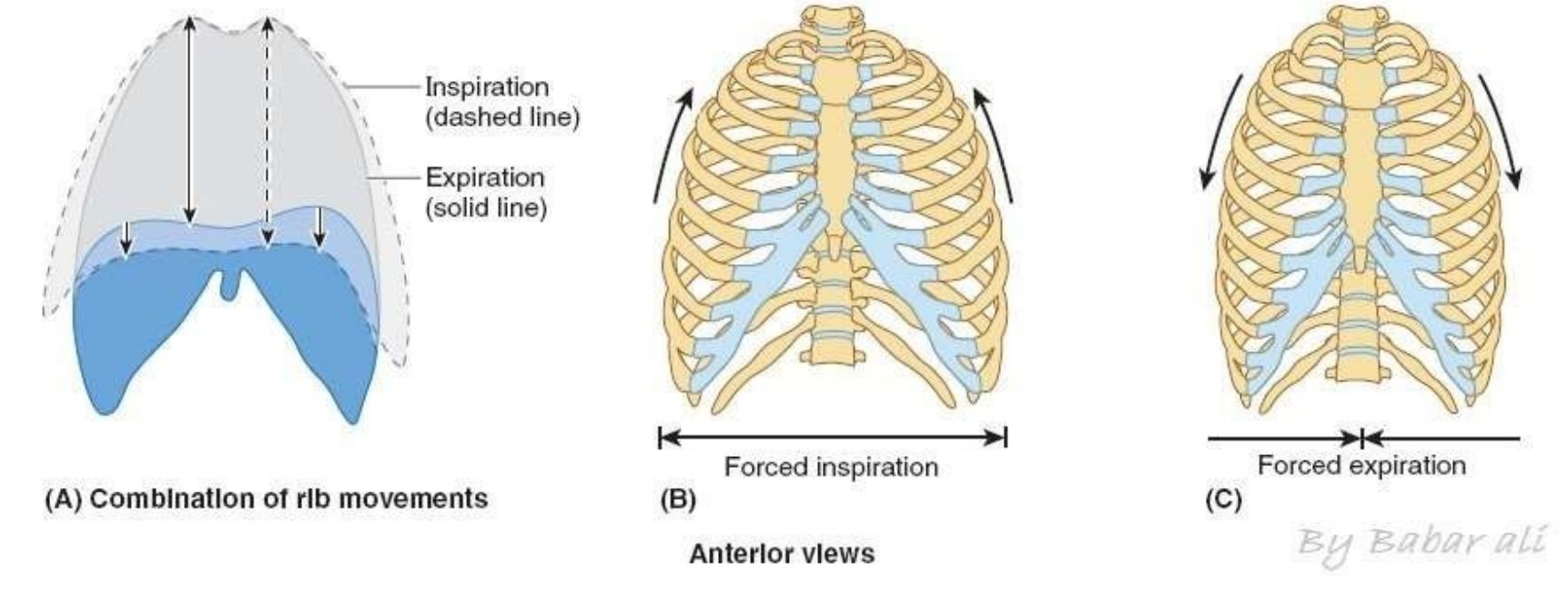
Actio

- During expiration these muscle contract to pull down (closed) the ribs to decrease surface area for lungs to expire.



Anteroposterior contraction

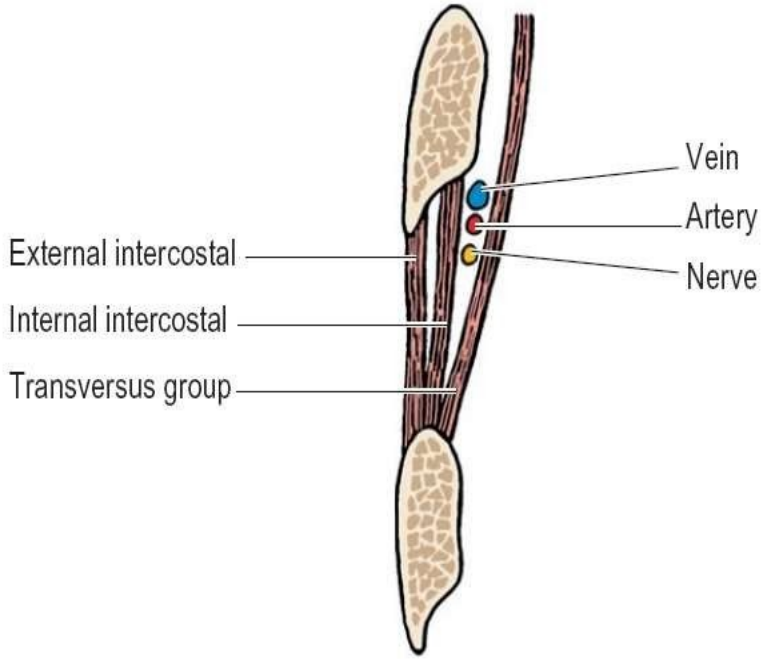
Schematic view

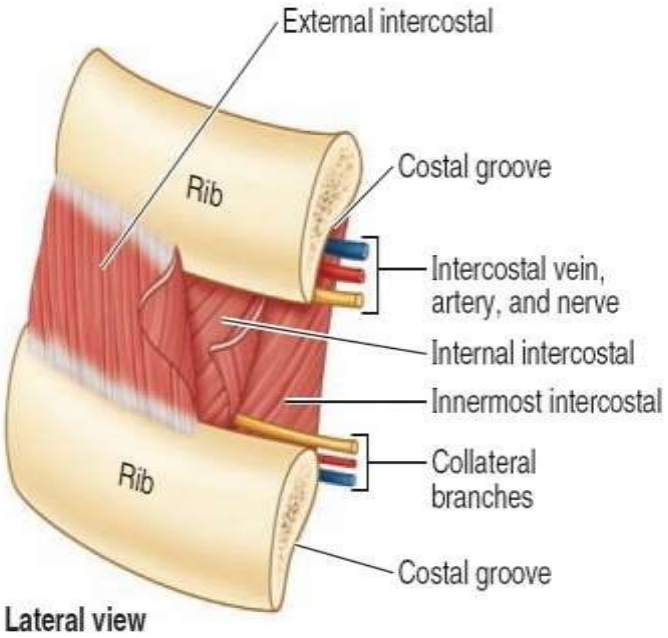


Innermost intercostal muscle

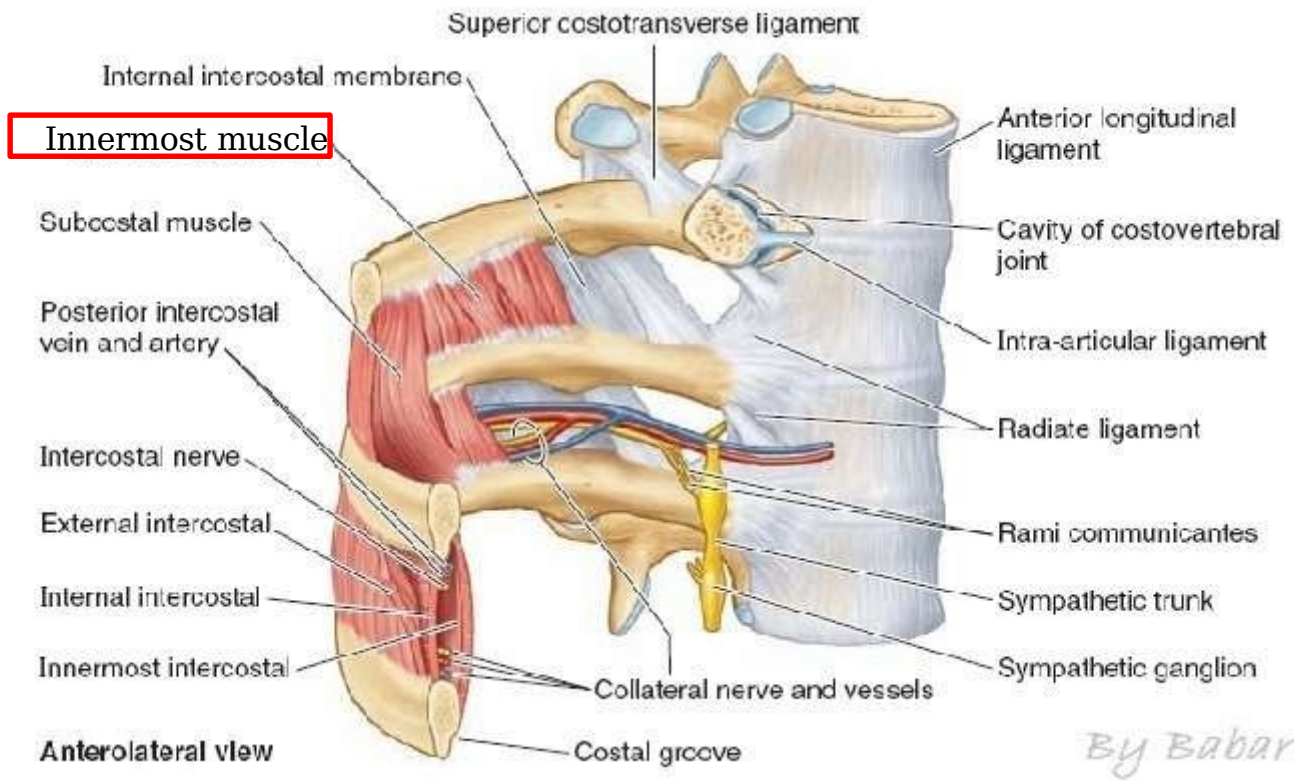
- Innermost intercostal – largely incomplete and consist of numerous slips of muscle tissue. They are individually named as the transversus thoracis (anteriorly), intracostal (laterally) and subcostalis (posteriorly) muscles. These are bound together by a continuous sheet of fascia. The innermost muscles are also separated from the parietal pleura by a further layer of fascia, the endothoracic fascia.

Innermost muscle





Innermost muscle



Schematic overview

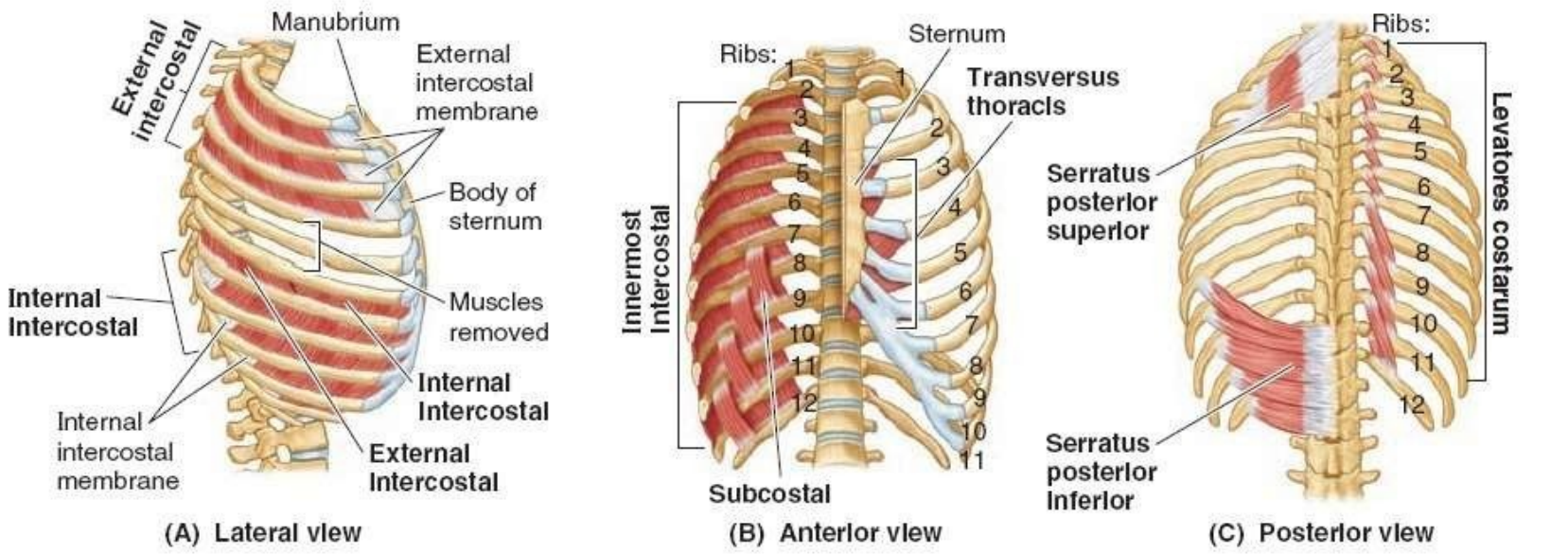


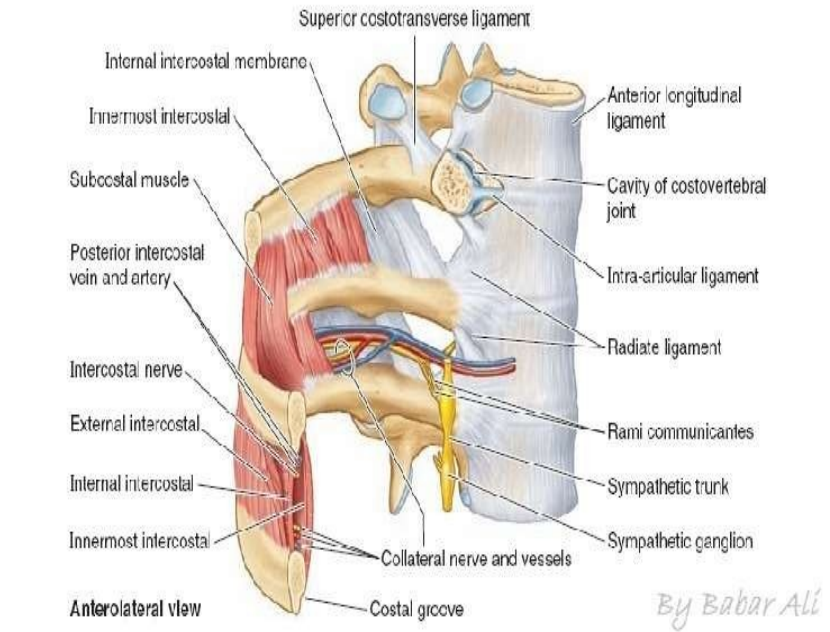
FIGURE **Muscles of thoracic wall. A.** External and internal intercostal muscles. **B.** Innermost intercostals, subcostal, and transversus thoracis muscles. **C.** Serratus posterior superior and inferior and levatores costarum muscles.

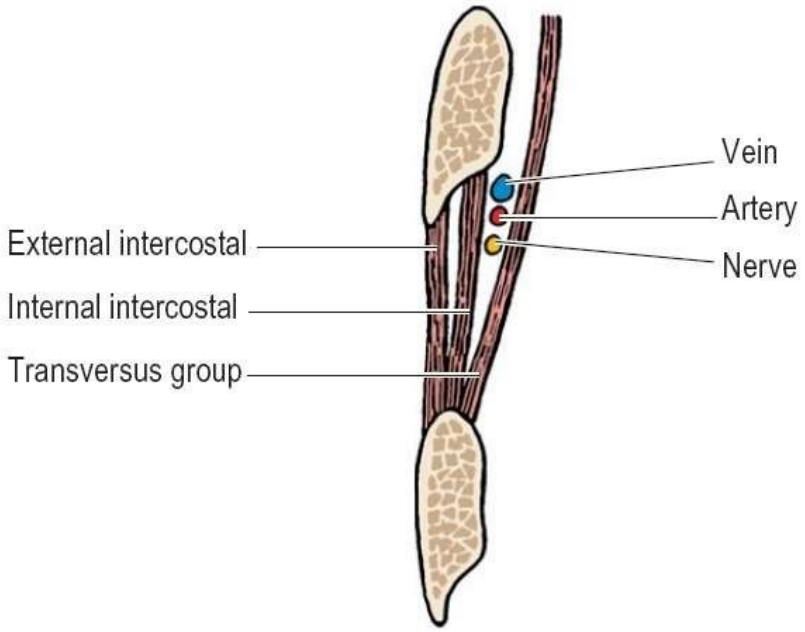
By Babar Ali

Neurovascular supply

- Neurovascular bundles – each consist of (from above down) a vein, an artery and a nerve i.e. (VAN). They lie between the innermost and internal intercostal muscle layers for almost their entire course:

Neurovascular supply



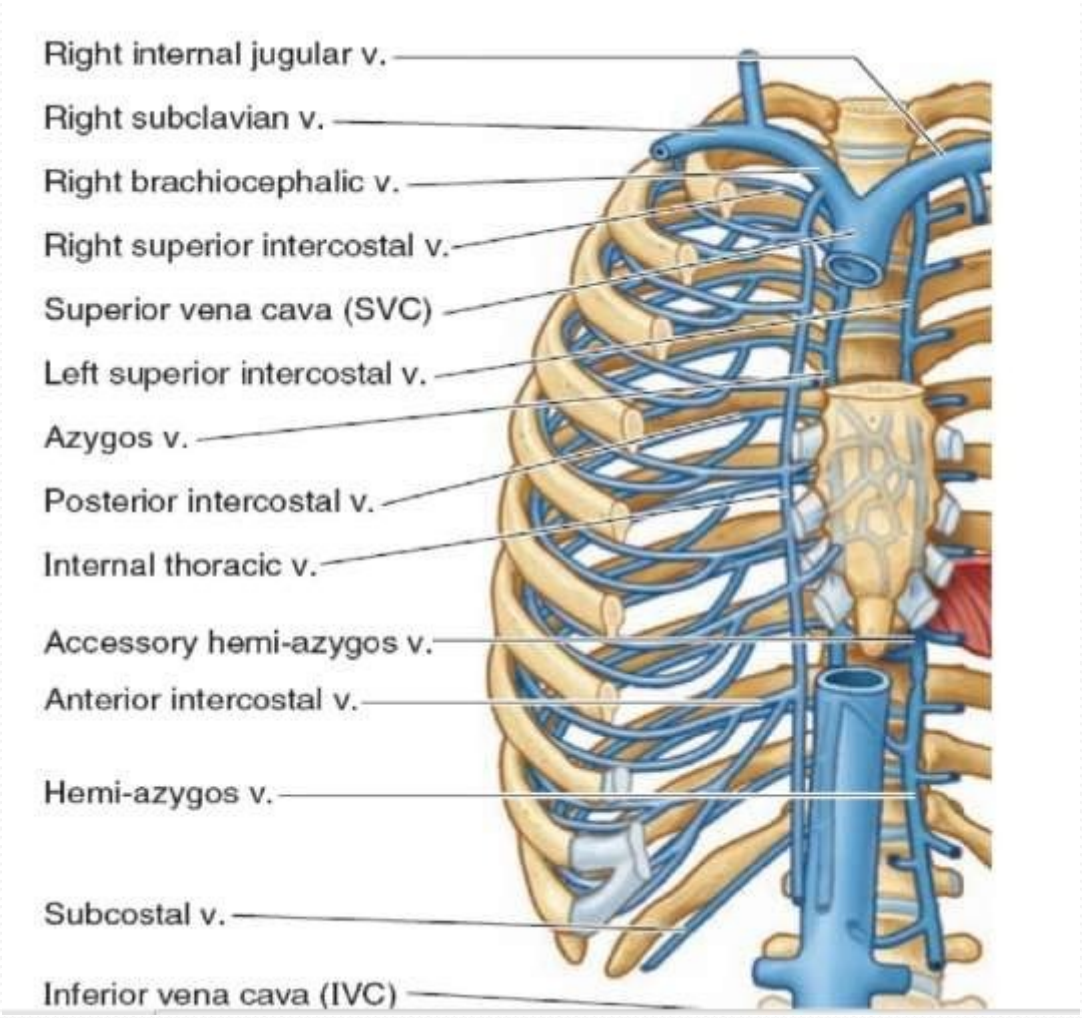


Venous drainage

- Veins – have complex terminations and consist of the posterior and anterior intercostal veins. The posterior veins eventually drain into the superior vena cava via the azygos and hemi-azygos systems, and the anterior veins drain into the musculophrenic vein (lower spaces), or the internal thoracic vein (upper spac



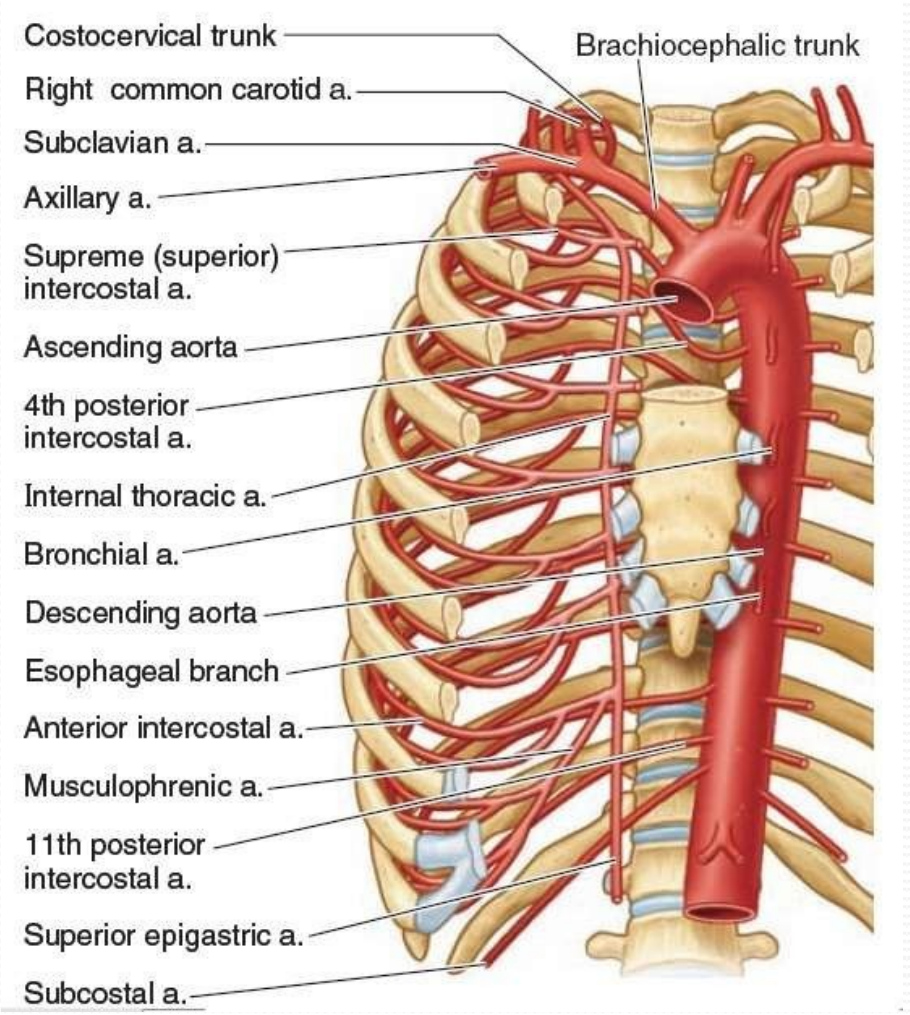
Venous drainage



Arterial supply

- Arteries – the posterior and anterior intercostal arteries. The posterior intercostal arteries arise directly from the thoracic aorta from T3 to T11, and from the superior intercostal artery (T1 and T2). The anterior intercostal arteries (T1–9) are derived from the two internal thoracic arteries, which themselves are branches of the subclavian artery, and run behind and just lateral to the sternal edge. The last two intercostal spaces are supplied by the posterior intercostal arteries only. The anterior intercostals eventually anastomose with the posterior intercostal arteries.

Arterial supply



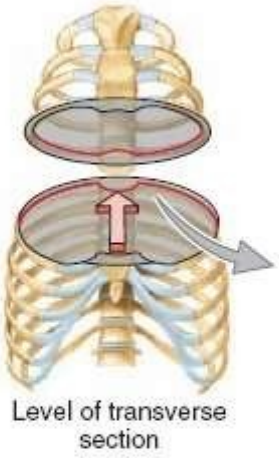
Nervous supply

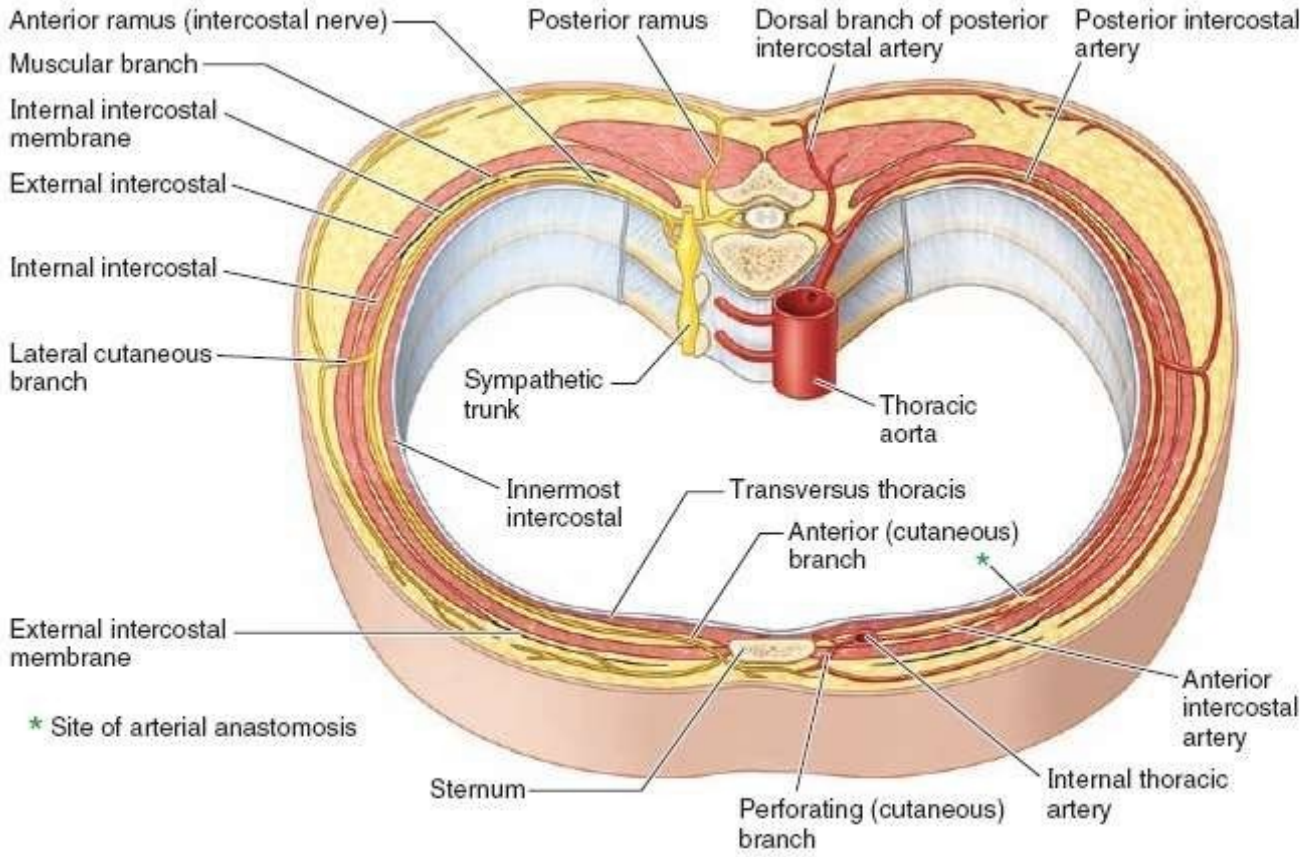
- Nerves – ventral (anterior) rami of the thoracic nerves from T1 to T11. The lower five nerves from T7 to T11 continue to supply the abdominal wall and maintain their position between the innermost and internal muscle layers (the internal oblique and transversus abdominus respectively). The typical intercostal nerve has the following branches:

Continue

- Rami communicantes to from the sympathetic trunk.
- Collateral to the intercostal muscles and pleura.
- Lateral cutaneous to the lateral wall skin and muscles.
- Anterior cutaneous to the anterior wall skin and muscles.

Schematic





Intercostal space, transverse section. This section shows nerves (*right side*) and arteries (*left side*).

Schematic

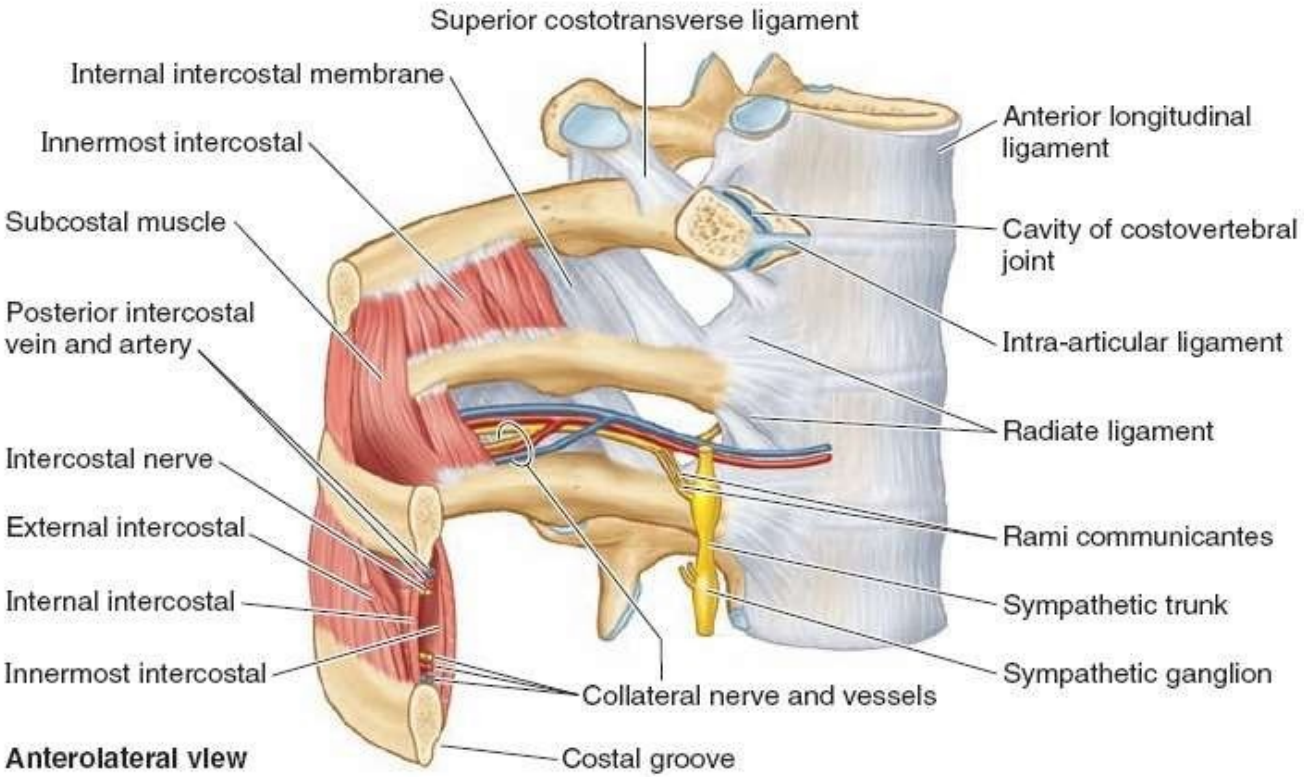


FIGURE **Posterior part of intercostal space.** Note the connection of the intercostal nerve to the sympathetic trunk by rami communicantes (communicating branches).

Importance of intercostals

- Help in breathing .
- Chest lead placement for ECG.
- Play import ant role in auscultation .
- Approachable way to percardial and pleural cavity.
- Prevent the internal tissue and organ from bulging out.
- Medium for internal organ of thoracic cavity during surgery.
- Prevent blowing in and out of air to maintain intrathoracic pressure .

Any question.....



Keep it……

"The man who try to do something and fail are infinitely better than those who try to do nothing and succeed"



Presented to: Mr. Sanaullah

Presented by: Babar Ali

Discipline: Cardiac Perfusion